Special right triangles

1 The side opposite the angle measuring 30°.

2

a $\frac{\sqrt{3}}{2}$ **b** $\frac{1}{2}$ **c** $\frac{1}{\sqrt{3}}$ **d** $\frac{1}{\sqrt{2}}$

3

a $\sqrt{3}$ **b** $x = 30^\circ$

c i $\frac{\sqrt{3}}{2}$ ii $\frac{1}{2}$ iii $\sqrt{3}$ **d** i $\frac{1}{2}$ ii $\frac{\sqrt{3}}{2}$ iii $\frac{1}{\sqrt{3}}$

Exact value trigonometric ratios

4

a 60° **b** 0.5

5

a 45° **b** 1

6

a 30° **b** 0.5

7

a 60° **b** 1.7

8

a 45° **b** 0.71

9

a 30° **b** 0.5

10

a 30° **b** 0.9

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11

a 30° **b** 0.58

12

a 60° **b** 1.73

13

a 60° **b** 0.9

14

a 45° **b** 60° **c** 60° **d** 30°

15

a $a = \sqrt{3}$ **b** $\theta = 60^\circ$

16

a 45° **b** 60°

17

a $x = \dfrac{\sqrt{30}}{2}$ **b** $h = 1$

c $x = 2$ **d** $P = 17$

e $w = 22$ **f** $x = 4\sqrt{6},\ y = 4\sqrt{3}$

g $u = 8,\ v = 4\sqrt{3}$ **h** $a = \dfrac{9\sqrt{3}}{2},\ b = \dfrac{9}{2}$

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Applications

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a 60°

b 30°

c x units

d $x\sqrt{3}$ units

e **i** False. The hypotenuse of a 30°- 60°- 90° triangle is twice the length of the shorter leg.

ii True

iii False. The longer leg of a 30°- 60°- 90° triangle is $\sqrt{3}$ times the length of the shorter leg.

iv True

19

a **i** $p = \frac{5\sqrt{3}}{3}$ **ii** $q = \frac{10\sqrt{3}}{3}$ **iii** $r = \frac{5}{2}$ **iv** $s = \frac{5\sqrt{3}}{2}$

b $\frac{15\sqrt{3}}{2} + \frac{5}{2}$

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